MILITARY SPECIFICATION

INDICATOR, LIQUID OXYGEN QUANTITY GMU-59/A

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid oxygen quantity indicator designated GMU-59/A. The indicator is of the capacitance type and indicates from 0 to 20 liters of liquid oxygen. The indicator cover glass is provided with a high efficiency antireflective coating.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-675	Coating of Glass Optical Elements (Anti-reflection)
MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Color

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-781	Reliability Tests, Exponential Distribution
MS33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form 0f

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. Indicators furnished under this specification shall be products which have passed the preproduction test as specified herein.

* 3.2 General specification. The requirements of MIL-I-25645 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction.

3.3.1 Configuration. The indicator shall consist of a single unit containing all of the equipment necessary to meet the performance requirements specified herein and in MIL-I-25645.

3.3.2 Capacitance. For design purposes, the capacitance for zero reading shall be 247.0 micromicrofarads (uuf).

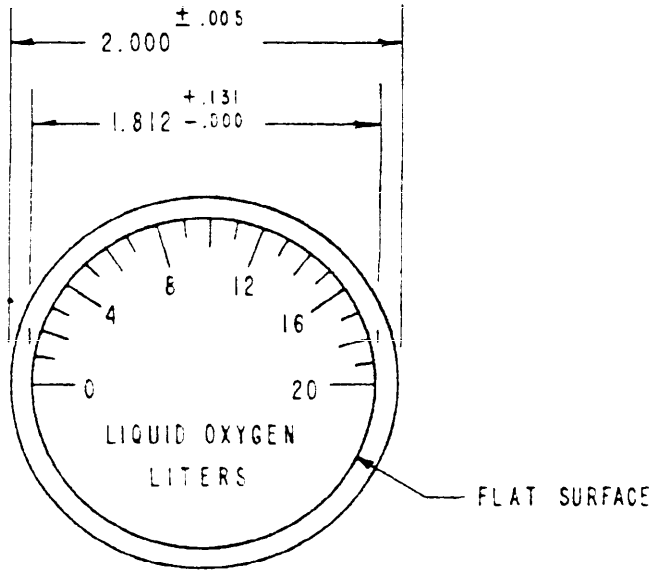
* 3.3.3 Integral lighting. The indicator shall have integral lighting conforming to MIL-L-27160 with the exception specified herein for a lamp circuit that uses a ground return. If the lamp circuit of the integral lighting uses a ground return, the dielectrict test specified in MIL-L-27160 shall be conducted prior to connecting the circuit to the ground.

3.4 Performance.

3.4.1 Accuracy. Accuracy of the indicator shall be in accordance with Table I when the indicator is tested as specified in MIL-I-25645. The tolerance for room-temperature readings is the allowable deviation from the nominal calculated values. All other tolerances are the allowable deviations from the actual room-temperature readings.

3.4.2 Adjustments. The zero calibration adjustment for the indicator shall be capable of adjusting the pointer to zero indication over an input capacitance range of at least 240.0 to 265.0 uuf. With the indicator adjusted to zero indication at 240.0 uuf input capacitance, the indicator shall be capable of calibration to full scale indication at 356.0 uuf input capacitance by means of the full calibration adjustment. With the indicator adjusted to zero indication at 265.0 uuf input capacitance, the indicator

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NUMERALS & LETTERS SHALL CONFORM TO MS33558

MARKINGS	HEIGHT OR LENGTH ± 0.010	WIDTH OF LINE OR GRADUATION ± 0.005	FINISH COLOR PER FED-STD-595
NUMERALS	0.140	0.025	
LIQUID OXYGEN	0.140	0.020	
MAJOR GRAD	0.188	0.031	
MINOR GRAD	0.125	0.015	37875
SUB MINOR	0.080	0.015	
LITERS	0.100	0.015	
BACKGROUND OF DIAL			37038

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES

FIGURE 1. Dial

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shall be capable of calibration to full scale indication at 381.0 uuf input capacitance by means of the full calibration adjustment.

3.5 Case. The length of the case shall not exceed 5.0 inches.

3.6 Dial. The dial for the indicator shall conform to Figure 1.

TABLE I. Tolerance for capacitance readings.

Test	Tolerance (uuf) Plus or Minus
Accuracy - room temperature	1.4
Low temperature operation	2.0
High temperature operation	2.0
Vibration	1.2
Power supply variation	1.2
Low temperature storage and operation	2.0
High temperature storage and operation	2.0
Salt spray	1.2
Cycling	1.2

3.7 Cover glass coating. The indicator cover glass shall be provided with a reflection-reducing coating which meets the requirements of MIL-C-675 except that the following reflectance tolerances shall apply:

Angle of Incidence	Wavelength (millimicrons)	Percent Reflectance
0°	450 - 675	0.6 maximum
0°	425 - 700	0.5 average
30°	450 - 625	1.0 maximum
30°	425 - 700	0.5 average

3.8 Electrical connector. The electrical connector shall be in accordance with Figure 2. The indicator shall be designed internally so that it will function properly when the mating connector is wired as shown on Figure 2.

3.9 Weight. The indicator shall weigh not more than 1.5 pounds.

* 3.10 Reclaimed materiel. The use of reclaimed materiels shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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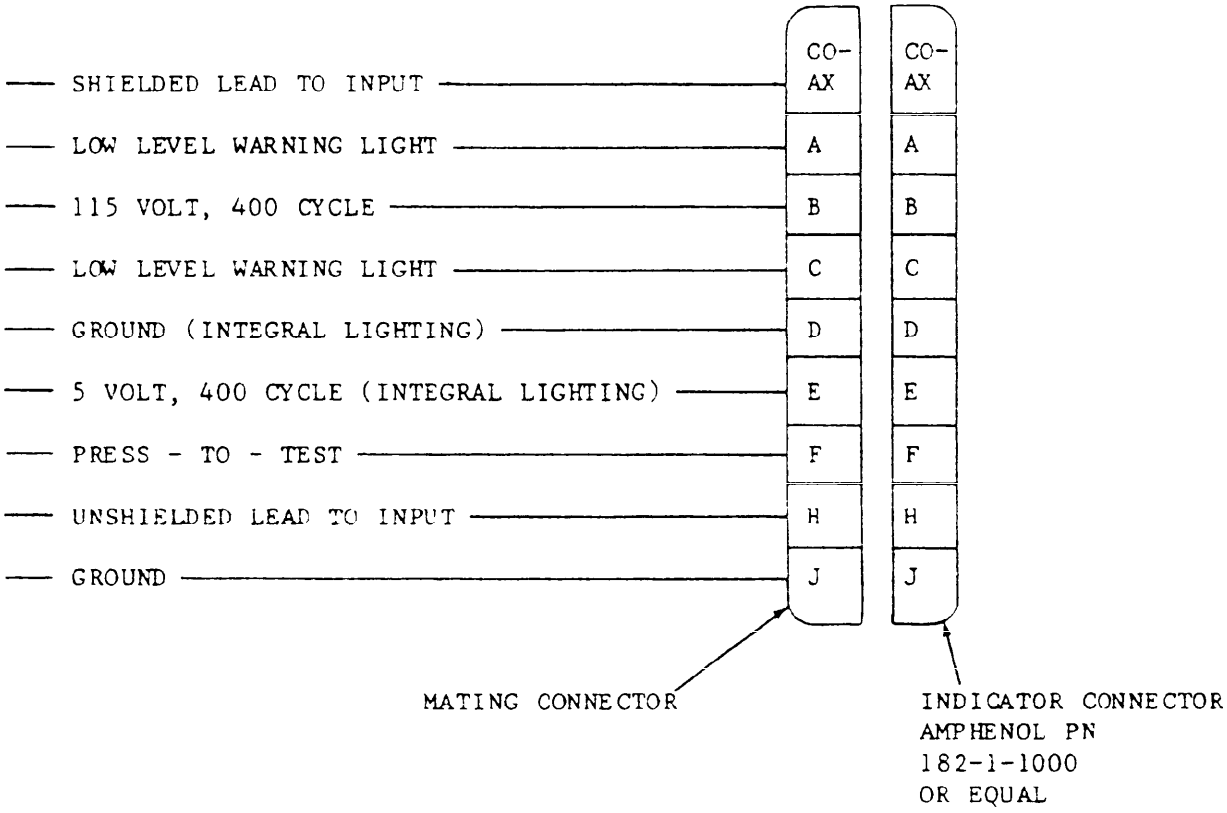


FIGURE 2. Connector wiring.

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General specification. Inspecting and testing of the indicators shall be as specified in MIL-I-25645 with the additional requirement for preproduction testing as specified in 4.3 of this specification.

* 4.3 Preproduction testing.

* 4.3.1 Preproduction test sample tested by the contractor. The contractor shall subject one indicator to the preproduction tests specified in 4.3.2.

4.3.2 Preproduction tests. Preproduction tests shall consist of all tests described under Test Methods of MIL-I-25645, and the additional test described in 4.4.1 of this specification.

* 4.3.3 Preproduction reliability tests. One preproduction sample of the equipment shall be tested for reliability and shall be continued through a longevity test in accordance with MIL-STD-781. Acceptance shall be judged on the basis of the accept-reject criteria of Test Plan I of MIL-STD-781. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

4.4 Test methods.

4.4.1 Cover glass reflectance. The reflectance of the indicator cover glass shall be measured and shall be within the tolerances specified in 3.7.

* 5. PACKAGING.

5.1 The indicator shall be prepared for delivery as specified in MIL-I-25645.

* 6. NOTES.

6.1 Intended use. The GMU-59/A indicator covered by this specification is intended for use in indicating the quantity of liquid oxygen remaining in converters in F-111 and other versions of this aircraft.

* 6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

*

a. Title, number and date of this specification

b. Levels of preparation for delivery required.

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* 6.2.2 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:
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a. Paragraph Number and Wording:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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